

Intro to DSP - MUSIC 320 //Homework 3 (100 Points)

Handed out on: October 20th, 2025 - 3:30 pm

Due on: November 3th, 2025 - 1:30 pm

Submission instructions

The homework must be submitted using Canvas's assignment submission system (in the Assignments tab). Select the associated link in that tab and proceed with your submission.

You must **turn in 2 items**:

- **A write-up (with all the instructed answers and figures)** in PDF format. Scanned and typed submission are both acceptable.
- **All your code and data files in a single compressed file** (zip, tar or tar.gz) containing the jupyter notebook with all the answers to the lab exercises and any other files requested in the homework instructions.

For your files, use the following **naming convention**: hwX.suid.type where X is the homework number and suid is your Stanford ID. For example, the write-up for homework 1 will be hw1_mab.pdf and the code files hw1_mab.zip/hw1_mab.tar.gz/hw1_mab.tar.

Theory Problems (60 pts.)

- (8 pts.) Even and Odd Functions: A sequence $x[n]$ is said to be *even* if $x[-n] = x[n]$, and *odd* if $x[-n] = -x[n]$, $n = 0, 1, 2, \dots, N-1$, where indexing is carried out modulo N . Find the even and odd parts of the following sequences:
 - (a) $[0, 1, 2, 1]$
 - (b) $[0, 1, 0, -1]$
 - (c) $[0, 1, 2, 3]$
 - (d) $x(n) = n^2$, $n = 0, 1, 2, \dots, N-1$
- (24 pts.) DFT Symmetry: If $Y(k)$ denotes the k th element of the length N DFT of y , show that:
 - (a) y real (i.e., $\text{im}\{y\} = 0$) $\iff Y(k) = Y[N-k]^*$ (DFT{real} is *Hermitian*)
 - (b) y imag (i.e., $\text{re}\{y\} = 0$) $\iff Y(k) = -Y[N-k]^*$ (*anti-Hermitian*)
 - (c) y even $\iff Y$ even
 - (d) y odd $\iff Y$ odd
 - (e) y real, even $\iff Y$ real, even
 - (f) y real, odd $\iff Y$ imag, odd
 - (g) y imag, even $\iff Y$ imag, even
 - (h) y imag, odd $\iff Y$ real, odd

3. (20pts.) DFT - Shift, FLIP, Linearity: Find the length $N = 8$ DFT

$$X_i(k), \quad k = 0, 1, \dots, 7,$$

for the following sequences:

- (a) $x_1 = [1, 0, 0, 0, 0, 0, 0, 0]$
 - (b) $x_2 = [0, 1, 0, 0, 0, 0, 0, 0]$
Use the shift operator to express $x_2(n)$ in terms of $x_1(n)$, and then compute the DFT of $x_2(n)$ in terms of the the DFT of $x_1(n)$.
 - (c) $x_3 = [0, 0, 0, 0, 0, 0, 0, 1]$
Use the flip operator to express $x_3(n)$ in terms of $x_2(n)$, and then compute the DFT of $x_3(n)$ in terms of the DFT of $x_2(n)$.
 - (d) $x_4 = [2, 1, 0, 0, 0, 0, 0, 1]$
Use linearity to express $x_4(n)$ in terms of $x_1(n)$, $x_2(n)$, and $x_3(n)$, and then compute the DFT of $x_4(n)$ in terms of $x_1(n)$, $x_2(n)$, and $x_3(n)$.
 - (e) Express $X_4(k)$ in terms of $X_1(k)$ and $X_2(k)$.
4. (8 pts.) Signal Metrics: For signals $y_1 = [1, 2, -3]$ and $y_2 = [1, j, 1 - j]$, find the
- (a) mean μ_y
 - (b) total signal energy $\mathcal{E}_y$
 - (c) average signal power $\mathcal{P}_y$
 - (d) sample variance σ_y^2

Lab Assignments (40 Points) Please see the attached Jupyter notebook.